

Smart Library Request Workflow in ServiceNow

Problem Statement

Traditional library systems often rely on manual processes or poorly integrated digital tools, leading to inefficiencies such as duplicate requests, lack of real-time book availability, and weak access control. Students may request already issued books, while librarians struggle to manage requests systematically. There is a need for a centralized, automated system that streamlines book requests, enforces access control, and improves overall library management.

Project Description

The Smart Library Request Workflow is a ServiceNow-based application designed to automate and optimize the process of requesting and managing library books. The system introduces role-based access control, structured data management using custom tables, and workflow automation using Flow Designer.

Students can request books, while librarians can review, approve, or reject requests. The system ensures that only available books can be requested and automatically updates book status when issued. UI policies enhance usability by preventing unwanted edits, and reports provide insights into frequently borrowed books.

This project demonstrates how ServiceNow can be used to build scalable, real-world enterprise applications with automation, data integrity, and user-friendly interfaces.

Skills and Technologies Used

- ServiceNow Platform Development
- Table Creation & Data Modeling
- Form Design & Field Configuration
- Reference Fields & Qualifiers
- Flow Designer (Workflow Automation)
- UI Policies & Client-Side Logic
- Access Control Lists (ACLs)
- Reporting & Data Visualization
- Role-Based Security Implementation

Project Initiation

The project begins with defining clear roles: Student and Librarian, ensuring proper separation of responsibilities. Custom tables such as "Book" and "Borrow Request" are created to store structured data.

Initial setup includes:

- Designing database schema
- Creating relationships between tables
- Defining fields and status choices
- Establishing role-based permissions

This phase lays the foundation for a secure and scalable system.

Execution Roadmap

The project is implemented in a step-by-step structured manner:

Milestone 1: Create Roles

Goal: Separate access → Students request, Librarians manage.

1. Open ServiceNow
2. Click on All → search for Roles.
3. Click on All → search for Roles.
4. Select User Administration → Roles.
5. Click on New.
6. Enter Role Name as librarian → click Submit.
7. Again, click on New.
8. Enter Role Name as student → click Submit.
9. Verify that both roles (*librarian* and *student*) are listed in the Roles module.

Milestone 2: Create Tables

Goal: Store Books and Requests

Activity 1 Create Book Table:

1. Open ServiceNow.
2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. Click on New.
5. Enter Label as Book → click Submit (this creates the table u_book).
6. Again, click on New.
7. Enter Label as Borrow Request → click Submit (this creates the table u_borrow_request).
8. Verify that both tables (Book and Borrow Request) are listed under the Tables module.

Milestone 3 : Add Fields to Book

Goal:Store book details

1. Open ServiceNow.
2. Click on All → search for Tables.
3. Select System Definition → Tables.
4. From the list, open the table Book (u_book).
5. Scroll down to the Columns (Dictionary) related list.
6. Click on New to add a new field.
7. Enter Column Label as *Title*, select **Type = String* → click Submit.
8. Again, click on New. Enter Column Label as *Author*, select **Type = String* → click Submit.
9. Again, click on New. Enter Column Label as *ISBN*, select **Type = String* → click Submit.
10. Again, click on New. Enter Column Label as *Status*, select **Type = Choice* → click Submit.
11. Open the Status field you just created.
12. In the Choices related list, click New and add values: *Available*, *Issued*, *Lost*.

13. Set the Default Value of Status = *Available*.
14. Click Update to save.
15. Verify that the Book form now has fields: *Title, Author, ISBN, Status*.

Milestone 4 Add Fields to Borrow Request :

Goal: Track request details

1. Open ServiceNow and click on All in the Application Navigator.
2. Search for Tables and select System Definition → Tables.
3. From the table list, open Borrow Request [u_borrow_request].
4. Scroll down to the Columns (Dictionary) related list.
5. Locate and open the field Book (Reference field to Book [u_book]).
6. In the field form, switch to Advanced View (click the gear  icon if Advanced View is not visible).
7. Scroll to the iplReference Qualifier section.
8. Select Simple as the Reference Qualifier type.
9. In the Reference Qualifier Condition, set the condition as: Status is Available.
10. Click Update to save the changes.

Milestone 5 Related List (Important for UI) :

Goal: See all requests inside a Book record

1. Open Flow Designer
2. Open ServiceNow.
3. Click on All → search for Flow Designer.
4. Select Flow Designer and click New → Flow.
5. Enter Name as Borrow Request Approval Flow → click Submit.

Milestone 6 Prevent Requesting Issued Books:

Goal: Only show Available books

1. Add Trigger (When to Run the Flow)
2. In the Flow Designer canvas, click + Add Trigger.

3. Select Created or Updated.
4. Choose Table = Borrow Request [u_borrow_request].
5. Add condition: Status = Requested. Click Done.

3. Add Action – Approval

1. Click on + Add Action.
2. Search for Ask for Approval.
3. Set Approvers = Role: librarian.
4. Approval conditions:
 - On Approve → continue flow.
 - On Reject → set Status = Rejected.
5. Click Done.

4. Add Action – Update Record (Issue Book)

1. Click on + Add Action.
2. Search for an Update Record.
3. Choose Table = Book [u_book].
4. Condition: Sys ID = [Current.Book](#) (the one from Borrow Request).
5. Set Status = Issued.
6. Click Done.

5. Add Action – Update Borrow Request Record

1. Click on + Add Action.
2. Search for an Update Record.
3. Choose Table = Borrow Request [u_borrow_request].
4. Condition: Sys ID = Current.
5. Set Status = Approved.
6. Click Done.

6. Add Action – Send Email (Notification)

1. Click on + Add Action.
2. Search for Send Email.
3. Set To = Requested By (Borrow Request → Requested By field).
4. Subject: Your Borrow Request has been Approved.
5. Body: The book [Book.Title] has been issued to you. Please collect it from the library.
6. Click Done

7. Save and Test Flow

- a. Click Save → Activate.
- b. Create a test Borrow Request with Status = Requested.
- c. Verify: Librarian gets approval → On approve, Book.Status = Issued, Borrow Request.Status = Approved, and student receives an email.

Milestone 7 Flow Designer (Automation)

Goal: When approved → Book becomes Issued

1. Open ServiceNow.
2. Click on All → search for UI Policies.
3. Select System UI → UI Policies.
4. Click on New.
5. Enter Table = Borrow Request [u_borrow_request].
6. Enter Short Description = Make Return Date mandatory when Issued.
7. Submit the record.
8. Reopen the created UI Policy record.
9. In the Conditions section, set: Status = Issued.
10. In the UI Policy Actions related list, click New.
11. Select Field = Return Date.
12. Set Mandatory = True.
13. Set Visible = True.
14. Submit the UI Policy Action.

Milestone 8 UI Policy

1. Open ServiceNow.
2. Click on All → search for UI Policies.
3. Select System UI → UI Policies.
4. Click on New.
5. Enter Table = Borrow Request [u_borrow_request].
6. Enter Short Description = Make Return Date mandatory when Issued.
7. Submit the record.
8. Reopen the created UI Policy record.
9. In the Conditions section, set: Status = Issued.
10. In the UI Policy Actions related list, click New.
11. Select Field = Return Date.
12. Set Mandatory = True.
13. Set Visible = True.
14. Submit the UI Policy Action.

Milestone 9 ACL

Goal: Restrict access properly

1. Open ServiceNow.
2. Click on All → search for Access Control.
3. Select System Security → Access Control (ACL).
4. Click New.
5. Select Table = Book [u_book].
6. Choose Operation = read.
7. Set Requires role = student, librarian.
8. Submit.
9. Again, click New.
10. Select Table = Book [u_book].
11. Choose Operation = write.
12. Set Requires role = librarian.
13. Submit.
14. Again, click New.
15. Select Table = Book [u_book].
16. Choose Operation = delete.
17. Set Requires role = librarian.
18. Submit.

Result: Students can only read books, librarians can read, write, and delete.

Milestone 10: Report

1. Open ServiceNow.
2. Click on All → search for Reports.
3. Select Reports → View / Run.
4. Click on New.
5. Enter Name = Most Borrowed Books.
6. Select Source Type = Table.

7. Select Table = Borrow Request [u_borrow_request].
8. Choose Type = Bar Chart (or Pie Chart as needed).
9. In the Group by field, select Book.
10. In the Aggregate field, select Count.
11. Add a filter: Status = Approved (to count only issued books).
12. Click Run to preview the report.
13. Adjust chart options (Title, Labels, Legend) as required.
14. Click Save. To publish,
15. click Sharing → Add roles (student, librarian).

Conclusion

The Smart Library Request Workflow successfully transforms a manual and inefficient library process into a fully automated, secure, and user-friendly system. By leveraging ServiceNow capabilities such as Flow Designer, ACLs, and UI policies, the project ensures accurate data handling, improved accessibility, and better decision-making through reports.

This project not only strengthens practical knowledge of ServiceNow development but also demonstrates the ability to design enterprise-level solutions—an essential skill for real-world IT and software roles.